Yuni Wu

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Summary

Data Engineer focused on trusted data products, production validation, and metadata-rich pipelines. Builds API, warehouse, and recommendation-data workflows that make operational data measurable, reconcilable, observable, and usable by downstream analytics and ML systems.

Experience

Graduate Research Assistant Sept 2025 – May 2026
CAIRO Lab, University of Colorado Boulder — Robotics and embodied AI research lab

- Engineered reproducible robot-learning datasets from ROS/rosbag telemetry by aligning camera imagery, motion-command supervision, checkpoints, and 97K+ Unitree Go1 state-action samples for downstream model validation.
- Improved autonomous waypoint prediction to 0.11 MAE and $R^2 = 0.62$ through behavior-cloning evaluation and model-selection metrics on the curated state-action datasets.

Machine Learning Engineer Intern May 2025 – Aug 2025
MyEdMaster LLC — AI-driven healthcare and education technology company

- Implemented repeatable data-validation workflows for a 22-class medical-image dataset with 50K+ imbalanced samples, supporting reproducible model evaluation and dataset-quality checks.
- Benchmarked evaluation-data quality and inference reliability through cross-validation, ROC-AUC, calibration, confusion-matrix analysis, and latency testing, including 0.92 F1-score and 5x lower latency from 0.15s to 0.03s.

Software Engineer June 2021 – Aug 2022
Shopee Co. Ltd. — Major Southeast Asian e-commerce platform

- Reduced manual QA workload by 80% by engineering Go API validation, SQL checks, and anomaly-detection workflows across 800+ production marketplace scenarios.
- Accelerated reporting speed by 78% by building regression monitoring pipelines that surfaced data inconsistencies earlier and shortened validation cycles.
- Cut query latency by 15% by optimizing distributed ETL workflows and GCP data-access patterns used in production data-quality and reporting paths.

Projects

NailAI — Cloud-Native ML Inference Metadata Platform [demo]

- Powered asynchronous image-processing workflows with FastAPI-backed Cloud Run services, Pub/Sub job routing, Cloud Storage asset handling, and BigQuery prediction metadata.
- Recorded 0 failures in a load test at 200 concurrent users with p95 latency of 6.8s through Pub/Sub job routing, retry handling, and backend/worker coordination.
- Improved inference observability by instrumenting request, job, and prediction metadata for monitoring, downstream analysis, and reliability investigation across services.

FitMatch — Product Size Recommendation Platform

- Standardized 1,205 apparel products into recommendation-ready catalog data by designing PostgreSQL schemas for products, size profiles, profile chart rows, and product-profile relationships.
- Achieved 24/24 passing validation cases by developing QA tooling, user-feedback capture, boundary regression tests, and Prisma-backed API validation.

Stock Intelligence Copilot — Traceable Data Product [demo]

- Unified market data, fundamentals, news, earnings, and portfolio signals into traceable multi-step analysis workflows with evidence bundles and per-tool error capture.
- Improved runtime traceability through FastAPI SSE streaming, structured execution schemas, route adapters, fallback-safe API presentation, and automated contract tests.

Technical Skills

Languages: Python, SQL, Go, R, TypeScript
Data Engineering: ETL/ELT workflows, data cleaning, data validation, schema validation, reconciliation, relational data modeling, metadata pipelines, data quality, query optimization, pandas, NumPy
Databases & Storage: BigQuery, PostgreSQL, MongoDB/pymongo, Cloud Storage, Prisma
Cloud & Backend: GCP Cloud Run, Pub/Sub, FastAPI, Flask, REST APIs, Docker, Kubernetes, asynchronous job orchestration, LangChain/LCEL, LangGraph, RAG
Testing & Reliability: anomaly detection, regression monitoring, observability metadata, fallback-safe APIs, automated regression testing, QA automation, Git, pytest/unittest, validation workflows

Education

University of Colorado Boulder — MS, Data Science | GPA: 3.76/4.0 May 2026
Soochow University — BS, Data Science | GPA: 3.7/4.0 June 2021